

SECTION B

7 (short)

(a) Define the bulk modulus. Starting with the generalisation of Hooke's Law relating linear stress and strain in 3D, show that the bulk modulus K of a linear elastic isotropic material is given by

$$K = \frac{E}{3(1-2\nu)}$$

where E is the Young's modulus and ν is the Poisson's ratio. Using this result, explain why it is improbable that a simple material could exist with $\nu > 0.5$. [5]

(b) A short cylindrical rod is made from a material with Young's modulus E and Poisson's ratio ν . The rod is compressed along its length so that the axial stress is $-\sigma$. The rod is constrained so that it cannot expand radially. Derive an expression for the axial strain in terms of E , ν and σ when the rod is compressed axially. [5]

8 (short)

(a) Sketch the unit cell for a Body Centred Cubic (BCC) material, showing the positions of the atoms in the cell. Identify an example close-packed direction and label the lattice parameter a . [4]

(b) The density of iron is 7870 kg m^{-3} . Using this value, calculate the lattice parameter a . Hence, find the atomic radius R . [6]